- *Pre-mission phase*: Mission intent, constraints, and ROE are loaded to the GCS; simulated plan verification runs on the digital twin; sensor calibrations and redundancy checks are performed automatically.
- *Launch and transit*: FCU and autonomy modules execute safe autonomous takeoff; federation protocols negotiate airspace access with deconfliction services; mesh/SATCOM links establish multi-layer connectivity.
- *On-station and cooperative operations*: Onboard perception fuses sensor inputs; local planners handle dynamic obstacles; distributed algorithms manage swarm roles (scout, loiter, strike, comms relay); compressed intelligence and metadata are transmitted rather than raw sensor streams to conserve bandwidth.
- *Dynamic re-tasking and escalation*: High-confidence detections or emergent targets trigger onboard classification and, depending on policy, either autonomous mitigation actions (e.g., evade) or operator alerts for clearance to engage. Explainable AI modules supply human supervisors with the rationale for autonomous decisions.
- *Anomaly handling and graceful degradation*: FDIR transitions the vehicle into safe loiter, return-to-home, or approved contingency patterns when failures are detected; redundant systems maintain minimum mission functionality.
- *Recovery and post-mission*: Secure data offload, automated health reports, and machine-learning model updates (federated learning) are synchronized with the cloud/enterprise systems to improve future performance.
- *Verification, certification, and governance*: Lifecycle integration includes model verification, SOTIF (safety of the intended functionality) analysis, and compliance with airspace and weapon